

NumPy

May 7, 2026

```
[24]: import datetime
x = datetime.datetime(2026, 5, 5)
print(x.strftime("%Y"))
```

2026

```
[19]: import datetime

x = datetime.datetime.now()
print(x)
```

2026-05-05 14:23:00.708269

```
[29]: import datetime
import datetime

now = datetime.datetime.now()
# Format date and time
current_date = now.strftime("%Y-%m-%d")
current_time = now.strftime("%H:%M:%S")

print("Today is:", current_date)
print("Current time is:", current_time)
```

Today is: 2026-05-05

Current time is: 14:30:52

```
[30]: import math
print(math.pi)
```

3.141592653589793

```
[31]: #Import math Library
import math

#Return factorial of a number
result= math.factorial(4)
print("Factorial: ", result)
```

Factorial: 24

```
[32]: import math

# Convert different degrees into radians
print(math.radians(180))
print(math.radians(90))
```

```
3.141592653589793
1.5707963267948966
```

```
[9]: import math

# Convert from radians to degrees:
print (math.degrees(3.141592653589793))
print (math.degrees(1.5707963267948966))
```

```
180.0
90.0
```

```
[33]: import math

# Return the sine of different values
print (math.sin(0.00))
print (math.sin(-1.23))

#To find the sine of degrees, it must first be converted into radians with the
↪math.radians() method
print(math.sin(math.radians(30)))
```

```
0.0
-0.9424888019316975
0.49999999999999994
```

```
[11]: import math

# Return the cosine of different numbers
print (math.cos(1.00))
print (math.cos(-1.5))

#To find the cosine of degrees, it must first be converted into radians with
↪the math.radians() method
print(math.cos(math.radians(30)))
```

```
0.5403023058681398
0.0707372016677029
0.8660254037844387
```

```
[12]: import math

# Return the tangent of different numbers
print (math.tan(90))
```

```

print (math.tan(-90))

#To find the tangent of degrees, it must first be converted into radians with
↳ the math.radians() method
print(math.tan(math.radians(30)))

```

```

-1.995200412208242
1.995200412208242
0.5773502691896257

```

```

[34]: import math

# Return the square root of different numbers
print (math.sqrt(9))
print (math.sqrt(25))

```

```

3.0
5.0

```

```

[35]: import math

# Round a number upward to its nearest integer
print(math.ceil(2.3))
print(math.ceil(5.7))
print(math.ceil(-4.6))

```

```

3
6
-4

```

```

[15]: import math

# Round numbers down to the nearest integer
print(math.floor(0.8))
print(math.floor(2.4))
print(math.floor(-6.8))

```

```

0
2
-7

```

```

[36]: # Exercise
#Task 1: Datetime Program (Short Code)

import datetime

now = datetime.datetime.now()

print("Year:", now.year)

```

```

print("Month:", now.month)
print("Day:", now.day)

print("Today is:", now.strftime("%A, %B %Y"))

```

Year: 2026
 Month: 5
 Day: 5
 Today is: Tuesday, May 2026

[17]: *# Task 2: Math Module Program (Short Code)*

```

import math

n = int(input("Enter a number: "))

print("Square Root:", math.sqrt(n))
print("Factorial:", math.factorial(n))
print("Sine:", math.sin(math.radians(n)))
print("Value of Pi:", math.pi)

```

Enter a number: 3
 Square Root: 1.7320508075688772
 Factorial: 6
 Sine: 0.052335956242943835
 Value of Pi: 3.141592653589793

[37]: `import numpy`
`arr = numpy.array([10, 20, 30, 40, 50 , "A", "B"])`
`print(arr)`
`print(arr.dtype)`

['10' '20' '30' '40' '50' 'A' 'B']
 <U11

[38]: *# Lecture 8-B: NumPy*

```

import numpy as np
arr = np.array([1, 2, 3, 4], ndmin=5)
print(arr)
print('number of dimensions :', arr.ndim)

```

[[[[[1 2 3 4]]]]]
 number of dimensions : 5

[39]: `import numpy as np`
`arr = np.array([[10,20,30,40], [50,60,70,80]])`
`print('3rd element on 1st dimension: ', arr[0, 2])`

3rd element on 1st dimension: 30

```
[40]: import numpy as np

arr = np.array([[1, 2, 3], [4, 5, 6]], [[7, 8, 9], [10, 11, 12]])
print(arr[0, 1, 2])
```

6

```
[59]: import numpy as np

arr = np.array([[1,2,3,4,5], [6,7,8,9,10]])
print('Last element from 2nd dim: ', arr[0, ::])
```

Last element from 2nd dim: [1 2 3 4 5]

```
[1]: import numpy as np
arr = np.array([1, 2, 3, 4, 5, 6, 7])
print(arr[1:5])
```

[2 3 4 5]

```
[15]: import numpy as np

arr = np.array([1, 2, 3, 4, 5])
x = arr.copy()
arr[0] = 50
print(arr)
print(x)
```

[50 2 3 4 5]

[1 2 3 4 5]

```
[2]: import numpy as np

arr = np.array([1, 2, 3, 4, 5])
x = arr.view()
arr[0] = 50
print(arr)
print(x)
```

[50 2 3 4 5]

[50 2 3 4 5]

```
[16]: import numpy as np

arr = np.array([1, 2, 3])
for x in arr:
    print(x)
```

1
2
3

```
[17]: import numpy as np

arr = np.array([[1, 2, 3], [4, 5, 6]])
for x in arr:
    print(x)
```

[1 2 3]
[4 5 6]

```
[18]: import numpy as np
arr = np.array([[1, 2], [3,4]])
for x in arr:
    for y in x:
        print(y)
```

1
2
3
4

```
[19]: import numpy as np
arr = np.array([[1, 2, 3], [4, 5, 6]], [[7, 8, 9], [10, 11, 12]])
for x in arr:
    print(x)
```

[[1 2 3]
 [4 5 6]]
[[7 8 9]
 [10 11 12]]

```
[20]: import numpy as np
arr = np.array([[1, 2, 3], [4, 5, 6]], [[7, 8, 9], [10, 11, 12]])
for x in arr:
    for y in x:
        for z in y:
            print(z)
```

1
2
3
4
5
6
7
8
9

10
11
12

[21]: *#Find the indexes where the value is 4:*

```
import numpy as np

arr = np.array([1, 2, 3, 4, 5, 4, 4])
x = np.where(arr == 4)
print(x)
```

(array([3, 5, 6], dtype=int64),)

```
[9]: import numpy as np
arr = np.array([3, 2, 0, 1])
print(np.sort(arr))
```

[0 1 2 3]

```
[10]: import numpy as np
arr = np.array(['c', 'a', 'b'])
arr2= np.sort(arr)[::-1]
print(arr2)
```

['c' 'b' 'a']

```
[24]: from numpy import random
x=random.randint(0,10, size=(3))
print(x)
```

[0 1 5]

```
[14]: from numpy import random
x = random.choice(['a', 'b', '7'])
print("Random Value: ", x)
```

Random Value: b

[]: